# Process Characterization Master Plan

**A-Mab Drug Substance — PCMP-001**

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

# Table of contents


Approvals . . . 3
Revision history . . . 3
Abbreviations . . . 4

**1 Purpose and scope** **5**
1.1 Changes in this revision . . . 7

**2 Stage 1 characterization strategy** **9**

**3 Critical quality attribute framework** **11**

**4 Risk-based prioritization** **13**

**5 Scale-down model strategy** **15**

**6 Common statistical approach** **19**
6.1 Designs . . . 19
6.2 Models, and what each model is for . . . 20
6.3 Model adequacy . . . 20
6.4 Proven acceptable ranges . . . 21
6.5 Capability, and the integrated model . . . 21

**7 Acceptance criteria framework** **23**
7.1 Scale-down model qualification . . . 23
7.2 Model adequacy . . . 23
7.3 Intermediate acceptance criteria . . . 24
7.4 Viral clearance claims . . . 24
7.5 Verification of the platform worst cases . . . 26
7.6 Proven acceptable ranges . . . 26
7.7 Parameter classification . . . 27
7.8 Method for fragments . . . 27
7.9 Capability and the probability of an out-of-specification drug substance . 29

**8 Register of characterization plans** **30**

**9 Deliverables and schedule** **31**

**10 Approvals** **34**

**References** **35**

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## Approvals

Table 2: Approval record (synthetic; signatures not applied).

| Role | Function | Name (synthetic) | Signature | Date |
|---|---|---|---|---|
| Author | Process Development / MSAT | — | — | — |
| Reviewer | Analytical Development | — | — | — |
| Reviewer | Manufacturing Science | — | — | — |
| Reviewer | Statistics | — | — | — |
| Approver | Quality Assurance | — | — | — |

## Revision history

| Version | Effective date | Description of change |
|---|---|---|
| 1.0 | 2025-08-04 | Initial issue |

| Version | Effective date | Description of change |
|---|---|---|
| 2.0 | 2026-09-30 | Re-analysis under INV-001: a-priori intermediate acceptance criteria, fragments by nrCE-SDS, two-sided capability for acidic variants, execution dates recorded (see INV-001) |

## Abbreviations

AEX, anion exchange chromatography; AMD, analytical method decision; AMV, analytical method validation; BLA, Biologics License Application; CEX, cation exchange chromatography; CPP, critical process parameter; Cpk, process capability index; CQA, critical quality attribute; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight species; IAC, intermediate acceptance criterion; KPP, key process parameter; LMW, low molecular weight species; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; nrCE-SDS, non-reduced capillary electrophoresis with sodium dodecyl sulfate; PAR, proven acceptable range; PPQ, process performance qualification; QA, Quality Assurance; RSM, response surface methodology; SEC, size exclusion chromatography; SOP, standard operating procedure; UF/DF, ultrafiltration and diafiltration; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukemia virus.

# 1 Purpose and scope

This plan governs the Stage 1 process characterization of the A-Mab drug substance process. It sits under the Process Transfer Plan PTP-001, which moves the process from the development site in Cambridge, MA, to the commercial drug substance facility in Grafton, WI. The characterization studies provide the evidence for the operating ranges and the control strategy that are transferred.

The plan covers the 8 steps of the drug substance train listed in Table 1.1, from the production bioreactor (Step 3) to ultrafiltration / diafiltration (Step 10). The seed expansion steps and the drug product process are outside its scope.

Table 1.1: The drug substance process train and the principal role of each step.

| Step | Unit operation | Principal role |
|---|---|---|
| 3 | Production Bioreactor | Forms the glycan, charge-variant and aggregate CQAs (design-space step) |
| 4 | Harvest and Clarification | Primary recovery and clarification; forms no product-quality CQA |
| 5 | Protein A Chromatography | Capture; principal HCP and DNA clearance; source of leached Protein A |
| 6 | Low-pH Viral Inactivation | Low-pH hold; inactivates enveloped virus (XMuLV credit) |
| 7 | Cation Exchange Chromatography | Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance |
| 8 | Anion Exchange Chromatography | Flow-through polish; clears HCP, DNA and virus (XMuLV and MVM credit) |
| 9 | Small-Virus Retentive Filtration | Dedicated small-virus removal; principal MVM clearance |
| 10 | Ultrafiltration / Diafiltration | Formulation / mass balance; delivers drug substance at target concentration |

This plan sets out the critical quality attribute (CQA) framework (Section 3), the risk basis (Section 4), the common approach to scale-down models (Section 5) and to statistics (Section 6), and the framework of acceptance criteria (Section 7) for the whole train. Each Process Characterization Plan (PCP-003 to PCP-010) applies this framework to one step. It adds what is specific to that step, such as the scale-down model, the factors and their ranges, and the sampling plan. Where a step plan departs from this plan, it should state the departure and the reason for it.

The documents of the package depend on each other in one direction. RA-001 sets the scope of each study. The PCP-00N plans the study, and the paired Process Characterization Report (PCR-00N) reports it. The Process Characterization Master Report PCMR-001 then consolidates the reports into the control strategy for the whole process. INV-001 is the investigation that led to this revision. Table 1.2 lists the package.

Table 1.2: Documents of the A-Mab characterization package.

| Document ID | Document class | Subject |
|---|---|---|
| PTP-001 | Process Transfer Plan | A-Mab Drug Substance |
| RA-001 | Pre-Characterization Process Risk Assessment | A-Mab Drug Substance |
| PCP-003 | Process Characterization Plan | Production Bioreactor (Step 3) |
| PCR-003 | Process Characterization Report | Production Bioreactor (Step 3) |
| PCP-004 | Process Characterization Plan | Harvest and Clarification (Step 4) |
| PCR-004 | Process Characterization Report | Harvest and Clarification (Step 4) |
| PCP-005 | Process Characterization Plan | Protein A Chromatography (Step 5) |
| PCR-005 | Process Characterization Report | Protein A Chromatography (Step 5) |
| PCP-006 | Process Characterization Plan | Low-pH Viral Inactivation (Step 6) |
| PCR-006 | Process Characterization Report | Low-pH Viral Inactivation (Step 6) |
| PCP-007 | Process Characterization Plan | Cation Exchange Chromatography (Step 7) |
| PCR-007 | Process Characterization Report | Cation Exchange Chromatography (Step 7) |
| PCP-008 | Process Characterization Plan | Anion Exchange Chromatography (Step 8) |
| PCR-008 | Process Characterization Report | Anion Exchange Chromatography (Step 8) |
| PCP-009 | Process Characterization Plan | Small-Virus Retentive Filtration (Step 9) |
| PCR-009 | Process Characterization Report | Small-Virus Retentive Filtration (Step 9) |
| PCP-010 | Process Characterization Plan | Ultrafiltration / Diafiltration (Step 10) |
| PCR-010 | Process Characterization Report | Ultrafiltration / Diafiltration (Step 10) |

| Document ID | Document class | Subject |
|---|---|---|
| PCMR-001 | Process Characterization Master Report | A-Mab Drug Substance |
| INV-001 | Investigation Report | Acceptance criteria for proven-acceptable-range analysis set after data review |

## 1.1 Changes in this revision

Version 1.0 of this plan was approved on 2025-08-04, before the first characterization study started on 2025-08-27. Version 2.0 replaces it and is issued under INV-001. QA opened the investigation on 2026-08-03 after its review of PCMR-001 Version 1.0 for BLA readiness, and closed it on 2026-09-18 with the classification major. INV-001 found that the criteria used to judge each step in the proven acceptable range (PAR) analysis were not in any approved protocol and had been set after the data were seen (INV-001 finding 1). Each criterion had been carried back from the drug substance limit through the nominal clearance of this campaign and then divided by a margin that differed by step (finding 2). Version 1.0 of this plan defined no criteria for step outputs.

This revision makes the following changes.

1. It defines intermediate acceptance criteria (IACs) for the step outputs on which a drug substance criterion depends, by one rule that uses no data from the studies it judges (Section 7.3). This is the corrective action for INV-001 findings 1 to 3.
2. It defines how each report verifies the platform worst cases that the criteria rely on (Section 7.5).
3. It sets non-reduced CE-SDS (nrCE-SDS) as the method for fragments at every step, as decided in AMD-001 on 2026-03-20 (Section 7.8, INV-001 finding 6).
4. It assesses capability for acidic variants against both ends of the criterion (Section 6.5, INV-001 finding 4).
5. It judges viral clearance on the mean log reduction and quality attributes on a single batch (Section 6.4). Version 1.0 judged every response on a single batch.
6. It records the execution window of every study and the calibration status of the equipment each study used (Section 5 and Section 9, INV-001 finding 5). The Version 1.0 title blocks of the package carried its compilation date, 2026-07-24, in place of the date each plan was approved. The revision history of this plan now gives the approval date.
7. It states campaign thresholds for model adequacy (Section 7.2) and for capability (Section 7.9). Version 1.0 left both to the individual study.

In Version 1.0 the cation exchange pool was judged against an aggregate limit of 1 % HMW. That limit was the drug substance criterion of 5 % divided by a margin of 5. No step after cation exchange removes aggregate. Under this plan the cation exchange pool therefore carries the drug substance criterion itself, 5 % (Table 7.1).

What moved at each step as a result is stated in the “Changes in this revision” section of its PCR-00N, from its own data.

# 2 Stage 1 characterization strategy

Process validation follows a lifecycle, and its first stage is process design (U.S. Food and Drug Administration 2011). In Stage 1 the commercial process is defined from the knowledge gained in development, and the studies that support it are documented. This campaign follows the enhanced approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012). Prior knowledge and risk assessment identify which parameters may affect which attributes. Characterization studies then measure those effects and bound them. The result is a set of normal operating ranges (NORs), PARs and parameter classes, from which the control strategy is built (CMC Biotech Working Group 2009).

A-Mab is a humanized IgG1 antibody developed on the Novacyte platform. The downstream train of Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, virus filtration and UF/DF was used for three earlier antibodies, X-Mab, Y-Mab and Z-Mab. Their history establishes the mechanism of each step and the range of performance to expect from it. Characterization of A-Mab therefore confirms and bounds known behaviour. The same history supplies the worst case of each downstream step, on which the intermediate acceptance criteria are built (Section 7.3).

Criticality is treated as a continuum, as described in ICH Q8(R2) (International Council for Harmonisation 2009). RA-001 ranks each parameter and assigns it to multivariate study, to univariate study or to no new study (Section 4). Parameters assigned to multivariate study are characterized by designed experiments in a qualified scale-down model. Such designs are planned for 6 steps, from the production bioreactor to virus filtration. The harvest (Step 4) and UF/DF (Step 10) steps are characterized by univariate and qualitative studies only, because RA-001 assigned none of their parameters to multivariate study. Their plans and reports are correspondingly shorter, and no designed experiment is planned for them.

For 31 of the 37 parameters the characterized range is wider than the NOR at both ends (Table 4.2). For 5 parameters (post-clarification turbidity, hold time, filtration volume (load), number of diavolumes and transmembrane pressure) the low end of the characterized range equals the low end of the NOR, and only the high end extends beyond it. For A-Mab concentration the NOR equals the characterized range. Where the range is wider, it shows how the process behaves near and beyond the edges of routine operation. It also gives the knowledge space within which a PAR can be claimed. A PAR is claimed only where the data support it, and never beyond the range studied.

Viral safety rests on modular and orthogonal clearance across low-pH inactivation, anion exchange and virus filtration (International Council for Harmonisation 2023). Each of these steps is studied in its own spiking model, and each step plan claims

only the log reduction of its own step. The cumulative claim is consolidated in PCMR-001.

The outputs of Stage 1 feed process performance qualification at the receiving site. PTP-001 plans 5 PPQ batches. The ranges and classes that this campaign supports are the ones those batches will be run against.

# 3 Critical quality attribute framework

The CQAs and their drug substance acceptance criteria are taken from the A-Mab case study (CMC Biotech Working Group 2009). Several of the criteria were set there from clinical experience with X-Mab, the closest platform antibody. They are held in the drug substance CQA register, and Table 3.1 lists all 11 attributes. Criticality was assigned with Tool #1 of the case study, as the product of impact and uncertainty. The levels run from very low (VL) to very high (VH), and the Tool #1 score stands beside each level.

Table 3.1: Drug substance CQA register, with acceptance criteria, criticality and the step at which each attribute is formed or set.

| CQA | Category | Acceptance | Criticality | Tool #1 | Formed or set at |
|---|---|---|---|---|---|
| Afucosylation | glycosylation | 2–13 % of glycans | H | 60 | Production Bioreactor |
| Galactosylation (total %Gal) | glycosylation | 10–40 % (G1+G2 equiv.) | H | 48 | Production Bioreactor |
| High Mannose | glycosylation | 3–10 % of glycans | VH | 80 | Production Bioreactor |
| Aggregates (HMW) | size variant | ≤ 5 % HMW (SEC) | H | 60 | Production Bioreactor |
| Acidic charge variants (deamidation) | charge variant | 18–40 % (CEX/icIEF) | VL | 4 | Production Bioreactor |
| Host Cell Protein (HCP) | process impurity | ≤ 100 ng/mg | M-H | 36 | Production Bioreactor |
| Residual DNA | process impurity | ≤ 0.001 ng/dose | VL | 6 | Production Bioreactor |
| Leached Protein A | process impurity | ≤ 5 ppm | L-M | 16 | Protein A Chromatography |
| Viral clearance — XMuLV (enveloped) | viral safety | ≥ 16.7 log10 (cumulative) | VH | 20 | Low-pH Viral Inactivation |
| Viral clearance — MVM (parvovirus) | viral safety | ≥ 8.6 log10 (cumulative) | VH | 20 | Anion Exchange Chromatography |
| Fragments (LMW, clipped species) | size variant | ≤ 2 % (nrCE-SDS) | L | 12 | Harvest and Clarification |

The last column of Table 3.1 names the step at which each attribute is formed or set. Most attributes are formed in the production bioreactor. The three glycan attributes are formed there, and no downstream step changes them. Protein A binds the Fc region at the interface of the CH2 and CH3 domains, so it captures every glycan variant alike. The two ion exchange steps do not separate glycan variants under platform conditions (CMC Biotech Working Group 2009). Leached Protein A enters the process at capture. Fragments are formed in culture and during the harvest hold, by proteases released from lysed cells that clip the antibody. They are assessed at harvest and at the downstream steps, and they are not an attribute of the production bioreactor designs (Table 7.3).

The two viral clearance attributes are cumulative. The register assigns each to one step, but each requirement is met by the sum of the claims of several steps (Section 7.4).

The attributes of very low criticality (acidic charge variants (deamidation) and residual DNA) remain in the register, and their criteria apply. A low criticality reduces the weight an attribute carries in RA-001. It does not remove the attribute from the acceptance framework.

The output of a step that removes an impurity, or that adds to a variant, is judged against an intermediate acceptance criterion derived from the drug substance criterion (Section 7.3).

# 4 Risk-based prioritization

RA-001 assessed every process parameter of the train before characterization began. Each parameter was ranked on its potential impact on the CQAs and on process performance, and on its potential to interact with other parameters. Parameters ranked high, or with a likely interaction, were assigned to multivariate study. Parameters ranked lower, whose range could be supported one factor at a time, were assigned to univariate study. RA-001 gives the scores and the rationale for each assignment, and they are not repeated here.

Table 4.1 shows the result across the train. Of the 37 parameters, 22 are studied in multivariate designs and 15 one factor at a time. All 6 parameters of harvest (Step 4) and UF/DF (Step 10) are studied univariately.

Table 4.1: Characterization scope per step, from RA-001.

| Step | Unit operation | Parameters | Multivariate | Univariate | Documents |
|---|---|---|---|---|---|
| 3 | Production Bioreactor | 9 | 5 | 4 | PCP-003 / PCR-003 |
| 4 | Harvest / Clarification | 3 | 0 | 3 | PCP-004 / PCR-004 |
| 5 | Protein A Chromatography | 6 | 4 | 2 | PCP-005 / PCR-005 |
| 6 | Low-pH Viral Inactivation | 4 | 3 | 1 | PCP-006 / PCR-006 |
| 7 | Cation Exchange Chromatography | 5 | 4 | 1 | PCP-007 / PCR-007 |
| 8 | Anion Exchange Chromatography | 5 | 4 | 1 | PCP-008 / PCR-008 |
| 9 | Small Virus Retentive Filtration | 2 | 2 | 0 | PCP-009 / PCR-009 |
| 10 | Ultrafiltration / Diafiltration (formulation) | 3 | 0 | 3 | PCP-010 / PCR-010 |

Table 4.2 lists every parameter with its set-point, NOR and characterized range. The class column gives the classification assigned in each PCR-00N under SOP-4001, by the rules in Section 7.7.

The range of a parameter studied univariately is supported by its own data, and its class is decided by the same rules as for a parameter studied in a design.

Table 4.2: Process parameters by step: set-point, ranges and classification.

| Step | Parameter | Unit | Set-point | NOR | | Characterized range | | Study | Class |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Low | High | Low | High | | |
| Production bioreactor | Culture pH | pH | 6.85 | 6.75 | 6.95 | 6.6 | 7.1 | multivariate | WC-CPP |
| | Culture temperature | °C | 35 | 34.5 | 35.5 | 34 | 36 | multivariate | WC-CPP |
| | Dissolved CO2 (pCO2) | mmHg | 100 | 70 | 130 | 40 | 160 | multivariate | WC-CPP |
| | Osmolality | mOsm | 400 | 375 | 425 | 360 | 440 | multivariate | WC-CPP |
| | Culture duration | days | 17 | 16 | 18 | 15 | 19 | multivariate | WC-CPP |
| | Dissolved oxygen | % sat | 50 | 40 | 60 | 30 | 70 | univariate[b)] | KPP |
| | Initial viable cell conc. | e6 cells/mL | 1 | 0.8 | 1.2 | 0.5 | 1.5 | univariate[b)] | KPP |
| | Basal medium concentration | X | 1.2 | 1 | 1.4 | 0.8 | 1.6 | univariate[b)] | GPP |
| | Nutrient feed-1 volume | % of working volume | 12 | 10 | 14 | 9 | 15 | univariate[b)] | KPP |
| Harvest | Centrifugation (rcf) | g | 9000 | 8000 | 10000 | 6000 | 12000 | univariate[b)] | KPP |
| | Depth filter load | L/m2 | 120 | 100 | 140 | 80 | 160 | univariate[b)] | KPP |
| | Post-clarification turbidity | NTU | 5 | 1 | 10 | 1 | 20 | univariate[b)] | GPP |
| Protein A | Protein load | g/L resin | 30 | 20 | 40 | 10 | 50 | multivariate | WC-CPP |
| | Elution buffer pH | pH | 3.55 | 3.4 | 3.7 | 3.2 | 3.9 | multivariate | WC-CPP |
| | Load flow rate | cm/hr | 200 | 150 | 250 | 100 | 300 | multivariate | KPP |
| | End of pool collect | CV | 2.6 | 2.3 | 2.9 | 2 | 3.2 | multivariate | KPP |
| | Operating temperature | °C | 22 | 18 | 26 | 15 | 30 | univariate[b)] | GPP |
| | Bed height | cm | 20 | 15 | 25 | 10 | 30 | univariate[b)] | GPP |
| Low-pH viral inactivation | Inactivation pH | pH | 3.5 | 3.4 | 3.6 | 3.2 | 4 | multivariate | CPP |
| | Hold time | min | 90 | 60 | 120 | 60 | 180 | multivariate | WC-CPP |
| | Temperature | °C | 21 | 19 | 23 | 15 | 25 | multivariate | WC-CPP |
| | A-Mab concentration | g/L | 20 | as characterized[a)] | | 5 | 35 | univariate[b)] | GPP |
| Cation exchange | Protein load | g/L resin | 25 | 18 | 32 | 10 | 40 | multivariate | WC-CPP |
| | Load/Wash conductivity | mS/cm | 5 | 4 | 6 | 3 | 7 | multivariate | WC-CPP |
| | Elution buffer pH | pH | 6 | 5.85 | 6.15 | 5.8 | 6.2 | multivariate | WC-CPP |
| | Elution stop collect | AU (OD280, descending) | 1 | 0.7 | 1.3 | 0.5 | 1.5 | multivariate | WC-CPP |
| | Elution flow rate | cm/hr | 200 | 150 | 250 | 100 | 300 | univariate[b)] | GPP |
| Anion exchange | Load pH | pH | 7.5 | 7.35 | 7.65 | 7.2 | 7.8 | multivariate | WC-CPP |
| | Equil/Wash-1 conductivity | mS/cm | 2.6 | 2.1 | 3.1 | 1.6 | 3.6 | multivariate | WC-CPP |
| | Load conductivity | mS/cm | 5.5 | 4 | 7 | 3 | 8 | multivariate | WC-CPP |
| | Protein load | g/L resin | 200 | 120 | 280 | 50 | 300 | multivariate | WC-CPP |
| | Operating flow rate | cm/hr | 300 | 200 | 400 | 150 | 450 | univariate[b)] | WC-CPP |
| Virus filtration | Filtration volume (load) | L/m2 | 90 | 50 | 105 | 50 | 140 | multivariate | WC-CPP |
| | Filtration pressure | psi | 13 | 10 | 20 | 8 | 30 | multivariate | WC-CPP |
| UF/DF | Number of diavolumes | DV | 8 | 7 | 9 | 7 | 10 | univariate[b)] | KPP |
| | Transmembrane pressure | psi | 12 | 10 | 14 | 10 | 15 | univariate[b)] | KPP |
| | Final DS concentration | g/L | 75 | 70 | 80 | 65 | 85 | univariate[b)] | KPP |

[a)] The normal operating range equals the characterized range.
[b)] Studied one factor at a time.

# 5 Scale-down model strategy

Every characterization study is run in a scale-down model. The model is qualified under SOP-1001 before its first characterization run. It is designed on the scaling principles of its unit operation. Parameters that do not depend on scale are held at their commercial values. Parameters that depend on scale are set so that the model matches the performance of the manufacturing scale.

For the chromatography steps the model keeps the bed height and the residence time of the manufacturing column. Load, wash and elution volumes are normalized to column volumes, and column efficiency is checked by plate count and peak asymmetry. For the production bioreactor the model is a stirred tank with equivalent mixing, aeration and process control. In each study, the parameters not under study are held at their commercial set-points. The designs vary culture pH, culture temperature, dissolved CO2 (pCO2), osmolality and culture duration, and univariate studies vary dissolved oxygen, initial viable cell conc., basal medium concentration and nutrient feed-1 volume. Agitation and gas flow are set to match the oxygen transfer and carbon dioxide stripping of the commercial vessel. For virus filtration the model uses the same filter type as manufacturing, and load is scaled by membrane area. For UF/DF, membrane loading and cross flow are scaled by membrane area.

Qualification compares the scale-down model, run at set-point, with manufacturing-scale data for the CQAs that have a criterion at the step (Table 7.1) and for step yield. Each PCP-00N defines that comparison and its criterion (Section 7.1). A model qualified for one output is not by that fact qualified for another. Where a model is not representative for an output, the PCP-00N should say so and should state how that output is characterized instead.

Viral clearance is studied in a spiking model of each claimed step, because virus cannot be introduced at manufacturing scale. The spiking model follows the same scaling principles as the characterization model of the step (International Council for Harmonisation 2023).

The calibration of 3 scale-down systems bears directly on the results. Table 5.1 lists them with the studies that used them. Every study window closed before the next calibration due date of the equipment it used. Every study that the Version 1.0 reports analysed closed before those reports were approved on 2026-07-24. The last of them ended on 2026-05-15. One supplementary qualification study, S-PA-SDM2 (2026-09-07 to 2026-09-11), ran afterwards, during INV-001, under DEV-005-03.

Table 5.1: Instrumented scale-down systems, the studies that used them, and their calibration.

| Equipment | Description | Study | Study window | Next calibration due |
|---|---|---|---|---|
| EQ-BRX-205 | production scale-down bioreactor pCO2 probe | S-BRX-SDM | 2025-08-27 to 2025-09-05 | 2026-06-14 |
| EQ-BRX-205 | production scale-down bioreactor pCO2 probe | S-BRX-UNI | 2025-10-06 to 2025-10-17 | 2026-06-14 |
| EQ-TFF-142 | ultrafiltration / diafiltration TFF skid | S-UFDF-SDM | 2026-02-25 to 2026-03-06 | 2026-04-22 |
| EQ-BRX-205 | production scale-down bioreactor pCO2 probe | S-BRX-SCR | 2025-09-08 to 2025-10-03 | 2026-06-14 |
| EQ-BRX-205 | production scale-down bioreactor pCO2 probe | S-BRX-RSM | 2025-10-20 to 2025-12-12 | 2026-06-14 |
| EQ-CHR-118 | temperature-controlled chromatography chamber | S-CEX-RSM | 2026-03-02 to 2026-03-27 | 2026-05-30 |
| EQ-TFF-142 | ultrafiltration / diafiltration TFF skid | S-UFDF-UNI | 2026-03-09 to 2026-03-27 | 2026-04-22 |

Intermediate pools are held between steps, and the hold can change the product. The campaign includes 3 hold studies. The clarified harvest and the Protein A eluate (at elution pH) are each held at 2-8 °C and 21 °C, and the neutralized viral inactivation pool at 21 °C and 25 °C, for up to 7 days. The design of the viral inactivation pool study follows A-Mab Table 4.17 (CMC Biotech Working Group 2009). Each hold study is planned in the PCP-00N of the step that produces the pool.

Each response is measured by a validated method under its own SOP. Table 5.2 lists every controlled procedure and method validation cited in the package. The SOP and AMV numbers are placeholders.

Table 5.2: Controlled procedures and method validations cited in the characterization package.

| Reference | Title | Type |
|---|---|---|
| SOP-1001 | Qualification of Scale-Down Models | SOP |
| SOP-1002 | Design, Execution and Statistical Analysis of DoE Studies | SOP |

| Reference | Title | Type |
|---|---|---|
| SOP-2003 | Operation of the Production Bioreactor (Fed-Batch) | SOP |
| SOP-2004 | Cell-Culture Media and Feed Preparation | SOP |
| SOP-2005 | Harvest by Continuous Disk-Stack Centrifugation | SOP |
| SOP-2006 | Clarification by Depth and Sterile Filtration | SOP |
| SOP-2007 | Operation of the Protein A Capture Chromatography Step | SOP |
| SOP-2008 | Chromatography Resin Life-Cycle, Packing and Sanitization | SOP |
| SOP-2009 | Operation of the Low-pH Viral Inactivation Step | SOP |
| SOP-2010 | Operation of the Cation-Exchange Polishing Chromatography Step | SOP |
| SOP-2011 | Operation of the Anion-Exchange Polishing Chromatography Step | SOP |
| SOP-2012 | Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step | SOP |
| SOP-2013 | Operation of the Ultrafiltration / Diafiltration (Formulation) Step | SOP |
| SOP-3010 | Released N-Glycan Mapping by 2-AB HILIC-UPLC | SOP |
| SOP-3011 | Size-Variant Analysis by SEC-HPLC | SOP |
| SOP-3012 | Host-Cell-Protein Quantitation by ELISA | SOP |
| SOP-3013 | Charge-Variant Analysis by icIEF | SOP |
| SOP-3014 | Residual Host-Cell DNA by qPCR | SOP |
| SOP-3015 | Purity by Non-Reduced CE-SDS | SOP |
| SOP-4001 | Process-Parameter Classification and Control-Strategy Definition | SOP |
| AMV-3010 | N-Glycan Map (2-AB HILIC-UPLC) | Method validation |
| AMV-3011 | Size-Variants (SEC-HPLC) | Method validation |
| AMV-3012 | Host-Cell Protein ELISA | Method validation |
| AMV-3013 | Charge Variants (icIEF) | Method validation |
| AMV-3014 | Residual DNA (qPCR) | Method validation |

| Reference | Title | Type |
|---|---|---|
| AMV-3015 | Turbidity by Nephelometry (NTU) | Method validation |
| AMV-3016 | Leached Protein A by ELISA (ppm) | Method validation |
| AMV-3017 | XMuLV Infectivity Titre (TCID50) | Method validation |
| AMV-3018 | MVM Infectivity Titre (TCID50/qPCR) | Method validation |
| AMV-3019 | Protein Concentration by A280 (UV) | Method validation |
| AMV-3020 | Purity (non-reduced CE-SDS) | Method validation |

# 6 Common statistical approach

## 6.1 Designs

Each step with a multivariate scope is studied in two stages, under SOP-1002. A screening design comes first. It is a two-level factorial design, full or fractional as the number of factors requires, with replicated centre points. A response-surface design follows. It is a central composite design over the factors that screening showed, or RA-001 expected, to affect a CQA with a criterion at the step. It also has replicated centre points. Table 6.1 gives the structure of each design. Every design has at least 3 centre points. The centre points estimate pure error and allow a test for curvature and for lack of fit.

Table 6.1: Designed experiments per step. Scr., screening design. Runs are given with the number of centre points in brackets.

| Step | Unit operation | Scr. factors | Scr. runs | Scr. model terms | RSM factors | RSM runs | Responses |
|---|---|---|---|---|---|---|---|
| 3 | Production Bioreactor | 5 | 19 (3) | 16 | 4 | 28 (4) | Afucosylation, Galactosylation, High mannose, Acidic variants, Aggregates (HMW) |
| 5 | Protein A | 4 | 19 (3) | 11 | 4 | 28 (4) | Pool HCP (ng/mg), Step yield, Leached Protein A (ppm), Fragments (LMW, %, nrCE-SDS) |
| 6 | Low-pH Viral Inactivation | 3 | 11 (3) | 7 | 3 | 18 (4) | Aggregates (HMW, %), XMuLV LRF ($\log_{10}$), Acidic variants, Fragments (LMW, %, nrCE-SDS) |
| 7 | Cation Exchange | 4 | 19 (3) | 11 | 4 | 28 (4) | Aggregates (HMW, %), Pool HCP (ng/mg), Step yield, Fragments (LMW, %, nrCE-SDS) |
| 8 | Anion Exchange | 4 | 19 (3) | 11 | 4 | 28 (4) | Pool HCP (ng/mg), XMuLV LRF ($\log_{10}$), MVM LRF ($\log_{10}$), Step yield, Fragments (LMW, %, nrCE-SDS) |

| Step | Unit operation | Scr. factors | Scr. runs | Scr. model terms | RSM factors | RSM runs | Responses |
|---|---|---|---|---|---|---|---|
| 9 | Small-Virus Retentive Filtration | 2 | 7 (3) | 4 | 2 | 12 (4) | MVM LRF ($\log_{10}$), XMuLV LRF ($\log_{10}$), Step yield, Fragments (LMW, %, nrCE-SDS) |

Runs are executed in random order within each design. Where a run is lost or invalidated, the PCR-00N should report it and state whether the design was repaired or re-executed.

## 6.2 Models, and what each model is for

The screening data are fitted with a model of main effects and two-factor interactions in coded units. The response-surface data are fitted with a full quadratic model. Effects are reported as the change in the response over the coded range of the factor, and significance is judged at $\alpha = 0.05$.

The screening model identifies effects. Its centre points detect curvature but cannot attribute it to a factor, and the model still interpolates linearly between the corners of the design. In the production bioreactor and small-virus retentive filtration screening designs the interaction model also leaves only 1 degree of freedom for lack of fit (Table 6.1), so its fit statistics overstate how well it fits. The response-surface model is the predictive model. Every prediction, every PAR and every design space in the package comes from it. No report should claim prediction from a screening model.

A response-surface model predicts the mean response over the range studied. At the edge of an acceptable region defined on mean predictions, a single batch meets its criterion about half the time if variability is symmetrical. That is why the PAR analysis in Section 6.4 carries the model residual and the variation of the other parameters into its decision.

## 6.3 Model adequacy

Each fitted model is reported with $R^2$, adjusted $R^2$ and predicted $R^2$, the model F-test, and the lack-of-fit test against pure error from the centre points. Residuals are inspected against predicted values and on a normal probability plot. A model is used for prediction only if it meets the adequacy criteria in Section 7.2.

## 6.4 Proven acceptable ranges

A PAR is computed for each CQA that has a criterion at the step and for each factor of the response-surface model. Two analyses are used, and each is referred to by the name given here.

1. The set-point analysis holds the other factors at their set-points and scans the factor across its characterized range. The PAR is the contiguous range around the set-point over which the mean prediction meets the criterion.
2. The NOR-propagated analysis also scans the factor. At each of 81 grid points it draws 2,000 operating points for the other factors. Each factor is drawn from a normal distribution centred on its set-point, with a standard deviation of one sixth of the width of its NOR, and clipped to the characterized range. The PAR is the contiguous range around the set-point over which the decision statistic meets the criterion.

The decision statistic depends on the kind of attribute. For a quality attribute it is the 95 % predictive interval of a single batch, including the model residual, because a single batch is what must meet the criterion. For viral clearance it is the 2.5th percentile of the mean log reduction, without the residual. The platform claim of a step is stated as an average log reduction factor (CMC Biotech Working Group 2009, 131). Most of the residual of a viral clearance model is the variability of the infectivity assay, not of the process. The single batch statistic is still computed for viral clearance and reported for information.

The NOR-propagated PAR is the one that supports a claim. It is in general narrower than the set-point PAR, because the range must hold while the rest of the operating point moves within its NOR.

## 6.5 Capability, and the integrated model

Capability at commercial scale is estimated from an integrated model of the train. The model runs 2,000 simulated batches through every step in sequence. Each process parameter of each batch is drawn as in the NOR-propagated analysis (Section 6.4). The drug substance attributes of each batch are recorded.

Capability is reported as Cpk against the drug substance criterion. For a criterion with one limit, Cpk is computed against that limit. For a two-sided criterion it is the lower of the two one-sided indices. Acidic variants have a two-sided criterion (18 to 40 %), and their capability is assessed against both ends. The criteria for capability are in Section 7.9.

The same simulation gives the probability of an out-of-specification drug substance for each attribute. Two figures are reported, because a few thousand simulated batches cannot resolve a rare event. These are the count of simulated batches outside the criterion, and the tail probability of a normal distribution with the simulated mean and standard deviation, one- or two-sided as the criterion is. The intermediate acceptance

criteria judge each step separately, and the integrated model shows whether the steps together deliver the drug substance.

# 7 Acceptance criteria framework

Each PCP-00N applies the criteria below to its own step. They define when a scale-down model is qualified, a model adequate, a step output acceptable, a range proven and a parameter classified. Where a value depends on the step, this plan defines the form of the criterion and the step plan sets its value before the study is run.

## 7.1 Scale-down model qualification

A scale-down model is qualified for an output when its runs at set-point agree with manufacturing-scale data for that output. Each PCP-00N names the outputs, the manufacturing-scale data set, the statistical comparison and the acceptance margin before any characterization run. A difference that is consistent and explained by a known scale effect may be accepted if the PCP-00N justifies it and the PCR-00N carries it into the interpretation. A model that fails qualification for an output is not used for that output.

## 7.2 Model adequacy

A response-surface model is adequate for prediction when the following hold.

1. The model is significant at $\alpha = 0.05$.
2. Lack of fit against pure error is not significant at $\alpha = 0.05$.
3. Predicted $R^2$ falls short of adjusted $R^2$ by no more than 0.20. This is a limit on overfitting, because a model that fits its own runs much better than it predicts left-out runs has more terms than the data support.
4. The residuals show no pattern against predicted values or run order, and no single run dominates the fit.

A model that fails criterion 2 or 3 is either refitted with a justified change of terms or transformation, or it is not used to set a range.

Where no adequate model results, the PCR-00N should report the response descriptively, against the criterion, over the runs of the design.

## 7.3 Intermediate acceptance criteria

A drug substance criterion does not say what the output of an upstream step must meet. Protein A, for example, leaves host cell protein in its pool that cation exchange and anion exchange will remove. Judging that pool against the drug substance limit of 100 ng/mg would fail every acceptable run. Each step output is therefore judged against an intermediate acceptance criterion.

The rule is the following. Each drug substance criterion is carried backward through the train, step by step, using the worst case that each downstream step showed on the platform. For a step that removes an attribute, the worst case is the lowest clearance fold observed for X-Mab, Y-Mab and Z-Mab. For a step that adds to an attribute, it is the largest increase observed. No margin is applied. An attribute that no downstream step changes keeps its drug substance criterion. This is the case for the three glycan attributes.

For host cell protein the rule gives the Protein A pool a criterion of 16,000 ng/mg. The drug substance limit is carried back through the platform worst case of anion exchange and then of cation exchange, which together remove host cell protein 160-fold. The cation exchange pool is judged against the limit carried back through anion exchange alone, a 4.00-fold clearance.

Table 7.1 gives all 21 criteria across 6 steps, with the downstream worst case each one is derived from. In the tightened rows the fold shown is the value that replaced the platform worst case (Section 7.5). The source of each platform value is recorded in the configuration of the process model and in the verification table of each report.

In Table 7.1, 2 criteria are tightened values. They are the aggregate criteria upstream of cation exchange, and Section 7.5 explains how they arose. The criterion derived a priori for the production bioreactor was 9.30 % HMW, and the criterion in force is 6.83 %. For low-pH viral inactivation the values are 10.0 % and 7.53 %.

Applying these criteria to studies already executed is acceptable because the a priori derivation uses no data from the studies it judges. The drug substance criteria come from the CQA register. The worst cases come from the platform history of X-Mab, Y-Mab and Z-Mab. A-Mab data enter a criterion only through verification (Section 7.5). The tightened criteria in force entered this way. Their values come from the A-Mab cation exchange model, and each is stricter than its a priori value.

## 7.4 Viral clearance claims

Each step that clears virus is held to its platform claim. The claim is the log reduction that the step has delivered for that virus on the platform, with X-Mab, Y-Mab and Z-Mab. It is a property of the step and its mechanism, and it does not depend on the other steps. The sum of the claims is then checked against the cumulative requirement of the drug substance. Table 7.2 shows that check for each model virus.

Table 7.1: Intermediate acceptance criteria and their derivation from the drug-substance criteria.

| Attribute | Drug-substance criterion | Step output | Downstream worst case | | Intermediate acceptance criterion | Basis |
|---|---|---|---|---|---|---|
| | | | Clearance fold | Increase | | |
| Host cell protein (ng/mg) | ≤ 100 | Protein A | 160 | 0.00 | ≤ 16,000 | Back-calculated |
| | | Cation exchange | 4.00 | 0.00 | ≤ 400 | Back-calculated |
| | | Anion exchange | 1.00 | 0.00 | ≤ 100 | Drug substance[a)] |
| Aggregates, HMW (%) | ≤ 5.00 | Production bioreactor | 1.51 | 0.70 | ≤ 6.83 | Back-calculated, tightened[b)] |
| | | Low-pH viral inactivation | 1.51 | 0.00 | ≤ 7.53 | Back-calculated, tightened[b)] |
| | | Cation exchange | 1.00 | 0.00 | ≤ 5.00 | Drug substance[a)] |
| Fragments, LMW (%) | ≤ 2.00 | Protein A | 1.30 | 0.15 | ≤ 2.45 | Back-calculated |
| | | Low-pH viral inactivation | 1.30 | 0.00 | ≤ 2.60 | Back-calculated |
| | | Cation exchange | 1.00 | 0.00 | ≤ 2.00 | Drug substance[a)] |
| | | Anion exchange | 1.00 | 0.00 | ≤ 2.00 | Drug substance[a)] |
| Leached Protein A (ppm) | ≤ 5.00 | Protein A | 31.6 | 0.00 | ≤ 158 | Back-calculated |
| Acidic variants (%) | 18.0–40.0 | Production bioreactor | 1.00 | 1.00 | 18.0–39.0 | Back-calculated |
| | | Low-pH viral inactivation | 1.00 | 0.00 | 18.0–40.0 | Drug substance[a)] |
| Afucosylation (%) | 2.00–13.0 | Production bioreactor | 1.00 | 0.00 | 2.00–13.0 | Drug substance[a)] |
| Galactosylation (%) | 10.0–40.0 | Production bioreactor | 1.00 | 0.00 | 10.0–40.0 | Drug substance[a)] |
| High mannose (%) | 3.00–10.0 | Production bioreactor | 1.00 | 0.00 | 3.00–10.0 | Drug substance[a)] |
| XMuLV clearance (log10) | ≥ 16.7[d)] | Low-pH viral inactivation | not applicable | | ≥ 6.60 | Platform claim[c)] |
| | | Anion exchange | not applicable | | ≥ 5.50 | Platform claim[c)] |
| | | Virus filtration | not applicable | | ≥ 4.62 | Platform claim[c)] |
| MVM clearance (log10) | ≥ 8.60[d)] | Anion exchange | not applicable | | ≥ 4.00 | Platform claim[c)] |
| | | Virus filtration | not applicable | | ≥ 4.62 | Platform claim[c)] |

[a)] No downstream step changes the attribute, so the drug-substance criterion applies.
[b)] The response-surface model of a downstream step predicted, within its NOR, a worse value than the platform worst case. The A-Mab value replaced it.
[c)] Log reduction claimed for this step in the modular viral clearance claim.
[d)] Cumulative requirement across the steps that clear the virus.

Table 7.2: Platform claims per step (log10) and their sum against the cumulative requirement of the drug substance.

| Virus | Low-pH viral inactivation | Anion exchange | Virus filtration | Sum of claims | Cumulative requirement | Met |
|---|---|---|---|---|---|---|
| XMuLV (enveloped) | 6.6 | 5.5 | 4.62 | 16.72 | 16.7 | yes |
| MVM (parvovirus) | no claim | 4 | 4.62 | 8.62 | 8.6 | yes |

The claims for XMuLV sum to 16.72 log10 against a requirement of 16.7. For MVM they sum to 8.62 against 8.6. Both requirements are met by the claims alone, with little room above them. A step that falls short of its claim therefore cannot be offset by the surplus of another step. Each viral step must meet its own claim. Low-pH viral inactivation makes no claim for MVM, because a parvovirus has no lipid envelope for low pH to disrupt.

## 7.5 Verification of the platform worst cases

The criteria rest on assumptions about the downstream steps. Each report checks the assumptions that the criteria upstream of its step rely on. The check uses the response-surface model of the step. An assumption is confirmed when the model predicts, within the NOR, a clearance no lower than the platform worst case, or an increase no larger. The values observed at the edges of the characterized range are reported beside it for information.

Data may tighten a criterion and may never loosen it. Where a step does worse within its NOR than the platform worst case, the value the A-Mab model predicts replaces the platform value, and every criterion upstream of that step is recalculated. Where a step does better, the platform value stands.

Of the 9 assumptions, 7 can be checked against a response of the campaign designs. The other 2 concern leached Protein A, which is not a response of the cation exchange or anion exchange designs, and the platform value stands for them. Of those 7, 6 were confirmed and 1 was not. Within its NOR, cation exchange cleared aggregate 1.51-fold, against a platform worst case of 2.00-fold. That tightened the 2 aggregate criteria upstream of cation exchange (Section 7.3).

## 7.6 Proven acceptable ranges

A range of a parameter is proven acceptable for an attribute when the NOR-propagated analysis (Section 6.4) meets the criterion of that attribute across it. The

criterion is the IAC where Table 7.1 defines one, and the drug substance criterion otherwise. A parameter that affects several attributes has a PAR for each, and the range carried into the control strategy is their intersection. Where the set-point itself fails, the PCR-00N reports that no PAR exists for the attribute and states the consequence for the NOR.

A PAR is never extended beyond the characterized range. A PAR that reaches the edge of the characterized range is reported as reaching it, and no claim is made beyond it.

## 7.7 Parameter classification

Parameters are classified under SOP-4001 by the decision logic of the A-Mab case study (CMC Biotech Working Group 2009).

1. A critical process parameter (CPP) has an effect on a CQA that is significant and practically relevant within the characterized range, and a high risk of falling outside its acceptable range in operation.
2. A well-controlled critical process parameter (WC-CPP) has the same kind of effect on a CQA but a low risk of falling outside its acceptable range, because it is readily controlled and its PAR is wide against its NOR.
3. A key process parameter (KPP) affects process performance, such as yield or concentration, and has no effect on a CQA within its range.
4. A general process parameter (GPP) has no relevant effect on a CQA or on process performance within its range.

A significant effect that is too small to move the attribute appreciably within the PAR does not by itself make a parameter critical. The PCR-00N should state the size of the effect beside its significance and give the reason for each class. The classification at this revision counts 1 CPP, 20 WC-CPP, 10 KPP and 6 GPP, and is consolidated in PCMR-001.

## 7.8 Method for fragments

From 2026-03-20, fragments are measured by nrCE-SDS (AMV-3020, under SOP-3015) in every study at every step, as AMD-001 decided. One method across the train is needed because a fragment balance across steps cannot mix two methods. Screening designs executed before that date measured fragments by SEC (Table 7.3).

The decision followed the comparability study S-CEX-FRC (2026-03-02 to 2026-03-13), in which SEC (AMV-3011) and nrCE-SDS were run on the same samples. SEC read fragments high wherever the sample still carried host cell protein, because the part of the host cell protein that elutes in the low molecular weight window of SEC is counted as fragment. The excess can never exceed the host cell protein present. It was a few tenths of a percent in the Protein A and viral inactivation pools and close to zero from the cation exchange pool onward. From the cation exchange pool onward

Table 7.3: Methods used for size variants, by step and study.

| Step | Attribute | Method used in study | |
|---|---|---|---|
| | | Screening | Response surface |
| Production bioreactor | Aggregates, HMW | SEC | |
| | Fragments, LMW | not in scope[f] | |
| Harvest | Aggregates, HMW | no DoE[e] | |
| | Fragments, LMW | no DoE[e] | |
| Protein A | Aggregates, HMW | SEC | |
| | Fragments, LMW | SEC[a] | SEC[a], nrCE-SDS[b] |
| Low-pH viral inactivation | Aggregates, HMW | SEC | |
| | Fragments, LMW | SEC[a] | SEC[a], nrCE-SDS[b] |
| Cation exchange | Aggregates, HMW | SEC | |
| | Fragments, LMW | SEC | SEC, nrCE-SDS[c] |
| Anion exchange | Aggregates, HMW | SEC | |
| | Fragments, LMW | nrCE-SDS[d] | nrCE-SDS[d] |
| Virus filtration | Aggregates, HMW | SEC | |
| | Fragments, LMW | nrCE-SDS | nrCE-SDS |
| UF/DF | Aggregates, HMW | no DoE[e] | |
| | Fragments, LMW | no DoE[e] | |

[a] In this matrix part of the host cell protein elutes in the SEC low molecular weight window. Reported for information only (AMD-001).
[b] Retained response-surface samples, tested 2026-04-06 to 2026-04-17.
[c] Both methods on the same samples, the comparability study S-CEX-FRC.
[d] Re-executed design (DEV-008-01). The invalidated first execution measured fragments by SEC.
[e] Univariate characterization only.
[f] Fragments were not an attribute of this study.

the two methods agreed (INV-001 finding 6). SEC remains the method for aggregate (HMW), where no such interference applies.

Table 7.3 shows which method measured fragments in which study. Retained response-surface samples of Protein A and low-pH viral inactivation were re-tested by nrCE-SDS between 2026-04-06 and 2026-04-17. Screening samples were not retained. Their SEC fragment results are kept in the record but are not used for any decision. The re-executed anion exchange design ran after the decision and measured fragments by nrCE-SDS. Fragments were not an attribute of the production bioreactor designs.

## 7.9 Capability and the probability of an out-of-specification drug substance

Capability is estimated as in Section 6.5 and judged for each CQA in PCMR-001 against two values.

1. A Cpk of at least 1.33 is expected of a CQA in routine manufacture. This value follows common industry practice for a capable process.
2. A CQA with a Cpk below 1 needs an action in the control strategy. At that value the process spread, at three standard deviations on the limiting side, just reaches the criterion. The action may be a tighter NOR, an added in-process control or added testing. PCMR-001 should name the action.

A Cpk between the two values is acceptable. PCMR-001 should then state the side of the criterion that limits it and the parameters that drive it. For the probability of an out-of-specification drug substance, PCMR-001 should report the count of simulated batches outside each criterion and the tail probability, and should discuss every attribute with a simulated batch outside its criterion.

# 8 Register of characterization plans

Table 8.1 lists the step plans that instantiate this master plan, with the study type each one carries and its paired report.

Table 8.1: Process Characterization Plans under PCMP-001.

| Plan | Subject | Study type | Paired report |
|---|---|---|---|
| PCP-003 | Production Bioreactor (Step 3) | screening and response surface | PCR-003 |
| PCP-004 | Harvest and Clarification (Step 4) | univariate and qualitative | PCR-004 |
| PCP-005 | Protein A Chromatography (Step 5) | screening and response surface | PCR-005 |
| PCP-006 | Low-pH Viral Inactivation (Step 6) | screening and response surface | PCR-006 |
| PCP-007 | Cation Exchange Chromatography (Step 7) | screening and response surface | PCR-007 |
| PCP-008 | Anion Exchange Chromatography (Step 8) | screening and response surface | PCR-008 |
| PCP-009 | Small-Virus Retentive Filtration (Step 9) | screening and response surface | PCR-009 |
| PCP-010 | Ultrafiltration / Diafiltration (Step 10) | univariate and qualitative | PCR-010 |

Each step plan carries the same structure. That is the purpose and scope, the objectives, the prior knowledge and risk basis, the materials and methods, the study design, the acceptance and decision criteria, data management, roles, deliverables, and the risks and assumptions of the study. The criteria section of each plan states the criteria of Section 7 for its own outputs and adds the values that this plan leaves to the step.

# 9 Deliverables and schedule

The campaign delivers the following documents.

1. One PCP-00N and one PCR-00N for each of the 8 steps, each report paired with its plan.
2. PCMR-001, which consolidates the reports into the design spaces, the PARs, the parameter classes, the capability of the drug substance and the control strategy.
3. A ground-truth annex for each controlled document, built from the same data as the document.

A step plan is approved before the first characterization run of its step. A report is approved after the data of all its studies are complete. PCMR-001 is written last.

Table 9.1 records the execution window of every study, in order of start. The campaign ran from 2025-08-27 to 2026-09-11 across 34 studies. Some steps were studied in parallel, because their scale-down models are independent of each other. The first execution of the anion exchange design (S-AEX-SUP) was invalidated under DEV-008-01 and re-executed as S-AEX-SCR and S-AEX-RSM.

Table 9.1: Execution window of every characterization study.

| Study | Step | Study type | Start | End |
|---|---|---|---|---|
| S-BRX-SDM | Production Bioreactor | scale-down model qualification | 2025-08-27 | 2025-09-05 |
| S-BRX-SCR | Production Bioreactor | screening design | 2025-09-08 | 2025-10-03 |
| S-BRX-UNI | Production Bioreactor | univariate | 2025-10-06 | 2025-10-17 |
| S-BRX-RSM | Production Bioreactor | response-surface design | 2025-10-20 | 2025-12-12 |
| S-HAR-UNI | Harvest and Clarification | univariate | 2025-11-03 | 2025-11-21 |
| S-PA-SDM | Protein A Chromatography | scale-down model qualification | 2025-11-05 | 2025-11-14 |
| S-PA-SCR | Protein A Chromatography | screening design | 2025-11-17 | 2025-12-05 |
| S-HAR-HLD | Harvest and Clarification | hold study | 2025-11-24 | 2025-12-05 |
| S-PA-UNI | Protein A Chromatography | univariate | 2025-12-08 | 2025-12-12 |
| S-VI-SDM | Low-pH Viral Inactivation | scale-down model qualification | 2026-01-07 | 2026-01-16 |

| Study | Step | Study type | Start | End |
|---|---|---|---|---|
| S-PA-RSM | Protein A Chromatography | response-surface design | 2026-01-12 | 2026-02-06 |
| S-VI-SCR | Low-pH Viral Inactivation | screening design | 2026-01-19 | 2026-01-30 |
| S-PA-HLD | Protein A Chromatography | hold study | 2026-02-02 | 2026-02-16 |
| S-VI-RSM | Low-pH Viral Inactivation | response-surface design | 2026-02-02 | 2026-02-13 |
| S-VI-UNI | Low-pH Viral Inactivation | univariate | 2026-02-02 | 2026-02-06 |
| S-AEX-SDM | Anion Exchange Chromatography | scale-down model qualification | 2026-02-04 | 2026-02-13 |
| S-CEX-SDM | Cation Exchange Chromatography | scale-down model qualification | 2026-02-04 | 2026-02-13 |
| S-AEX-SUP | Anion Exchange Chromatography | first execution (invalidated) | 2026-02-16 | 2026-03-04 |
| S-CEX-SCR | Cation Exchange Chromatography | screening design | 2026-02-16 | 2026-02-27 |
| S-VI-HLD | Low-pH Viral Inactivation | hold study | 2026-02-16 | 2026-02-27 |
| S-UFDF-SDM | Ultrafiltration / Diafiltration | scale-down model qualification | 2026-02-25 | 2026-03-06 |
| S-CEX-FRC | Cation Exchange Chromatography | method comparability | 2026-03-02 | 2026-03-13 |
| S-CEX-RSM | Cation Exchange Chromatography | response-surface design | 2026-03-02 | 2026-03-27 |
| S-UFDF-UNI | Ultrafiltration / Diafiltration | univariate | 2026-03-09 | 2026-03-27 |
| S-CEX-UNI | Cation Exchange Chromatography | univariate | 2026-03-16 | 2026-03-20 |
| S-AEX-SCR | Anion Exchange Chromatography | screening design | 2026-03-23 | 2026-03-31 |
| S-AEX-RSM | Anion Exchange Chromatography | response-surface design | 2026-04-01 | 2026-04-10 |
| S-PA-RET | Protein A Chromatography | retest of retained samples | 2026-04-06 | 2026-04-17 |
| S-VI-RET | Low-pH Viral Inactivation | retest of retained samples | 2026-04-06 | 2026-04-17 |
| S-AEX-UNI | Anion Exchange Chromatography | univariate | 2026-04-13 | 2026-04-17 |
| S-VF-SDM | Small-Virus Retentive Filtration | scale-down model qualification | 2026-04-15 | 2026-04-24 |
| S-VF-DOE | Small-Virus Retentive Filtration | screening design | 2026-04-27 | 2026-05-05 |
| S-VF-RSM | Small-Virus Retentive Filtration | response-surface design | 2026-05-06 | 2026-05-15 |

| Study | Step | Study type | Start | End |
|---|---|---|---|---|
| S-PA-SDM2 | Protein A Chromatography | scale-down model qualification | 2026-09-07 | 2026-09-11 |

# 10 Approvals

This plan is approved by the functions listed in the approval record at the front of this document, and each step plan is approved by the same functions before its studies start.

Table 10.1: Responsibilities of the approving functions.

| Function | Responsibility for this plan |
|---|---|
| Process Development / MSAT | Author. Owns the strategy, the designs and the acceptance framework |
| Analytical Development | Methods, method validations and the fragment method decision |
| Manufacturing Science | Scale-down model qualification and transfer to the receiving site |
| Statistics | Designs, models, PAR analysis and the integrated model |
| Quality Assurance | Approval, and closure of the actions from INV-001 |

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